

Multi-Scale Frequency Pyramid Fusion with Hierarchical Attention for Explainable Medical Image Classification and Interpretability

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by

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Evaluation Sheet

Title of the Project:

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Declaration

We, **Naga Kushal Vankadara, Yalamanchili Venusai** with Roll No: **122CS0015, 122CS0021**, hereby declare that the material presented in the Project Report titled **Multi-Scale Frequency Pyramid Fusion with Hierarchical Attention for Explainable Medical Image Classification and Interpretability** represents original work carried out by me in the **Department of Computer Science and Engineering** at the **Indian Institute of Information Technology Design and Manufacturing, Kurnool** during the years **2025 - 2026**. With our signatures, we certify that:

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In my capacity as supervisor of the above-mentioned work, I certify that the work presented in this Report is carried out under my supervision, and is worthy of consideration for the requirements of B.Tech. Project work.

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Abstract

Radiological reading remains a cornerstone of clinical decision-making, and modalities such as magnetic resonance imaging (MRI) and chest X-rays are routinely used by clinicians during triage and follow-up. Convolutional neural networks (CNNs) have become the default tool for automating parts of this process, yet almost all popular architectures operate directly on the pixel grid, and in doing so they tend to under-represent the high-frequency cues that differentiate small or low-contrast pathologies. The work reported in this thesis pursues an alternative route: we treat the spectrum of the image as a first-class input, and we do so in two stages that deliberately share the same backbone family so that the contributions of each stage can be isolated. In the first stage we introduce the **Dual-Frequency Feature Fusion (DFFF)** framework; for every RGB input we apply a per-channel Fast Fourier Transform (FFT) and retain three derived channels per colour band — the log-magnitude together with the cosine and sine of the phase — which gives a 9-channel spectral tensor that keeps both the energy distribution and the spatial structure of the original image. A CNN whose first convolution is widened to accept the 9-channel tensor is then paired with a dual-domain fusion block that blends spatial and spectral activations through learnable channel attention rather than a hand-picked weighting. The second stage generalises this design into the **Multi-Scale Frequency Pyramid Fusion (MSPF)** architecture, which is organised around a **Hierarchical Frequency-Spatial Attention (HFSA)** mechanism. HFSA is built from three cooperating pieces: lightweight Frequency Attention Gates (FAG) are attached at Layer 2 and Layer 3, where the FFT magnitude of the activations modulates the signal in a residual-style gating pattern; the DFFF block is retained at Layer 4, where full dual-domain fusion is applied to the most abstract features; and a Feature Pyramid Network (FPN) ties these scales together using softmax-weighted logits that are themselves learnt end-to-end. By separating these roles we let the network exploit fine edge and texture cues at high resolution while still reasoning about the coarser semantic structure captured deeper in the backbone. Interpretability is handled through a frequency-domain CAM variant, so that the regions on which each diagnosis hinges can be visualised directly on the original anatomical image.

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Abbreviation	Full Form
AMP	Automatic Mixed Precision
BN	Batch Normalization
CAM	Class Activation Mapping
CNN	Convolutional Neural Network
DFFF	Dual-Frequency Feature Fusion
DFT	Discrete Fourier Transform
FAG	Frequency Attention Gate
FFT	Fast Fourier Transform
FPN	Feature Pyramid Network
GAP	Global Average Pooling
HFSA	Hierarchical Frequency-Spatial Attention
IFFT	Inverse Fast Fourier Transform
LIME	Local Interpretable Model-agnostic Explanations
MRI	Magnetic Resonance Imaging
MSPF	Multi-Scale Frequency Pyramid Fusion
ReLU	Rectified Linear Unit
RGB	Red, Green, Blue
SE	Squeeze-and-Excitation
SHAP	SHapley Additive exPlanations
XAI	Explainable Artificial Intelligence

Symbol	Description
$F(u, v)$	2D Discrete Fourier Transform of an image
$M_c(u, v)$	Magnitude spectrum for channel c
$\Phi_c(u, v)$	Phase spectrum for channel c
$L_c(u, v)$	Log-compressed magnitude spectrum
X_{freq}	9-channel frequency-domain tensor
α, β	Learnable fusion weights
ϵ	Numerical stability constant / Label smoothing parameter
η	Learning rate
σ	Sigmoid activation function
δ	ReLU activation function
$\mathbf{W}$	Weight matrix
$\odot$	Element-wise (Hadamard) product
$\oplus$	Element-wise addition

*Dedicated to our parents and teachers, whose unwavering support,
encouragement, and belief in me have been the foundation of all our
achievements.*

Chapter 1

Introduction

1.1 Background and Motivation

The classification of medical images plays a significant role in the modern healthcare system as it assists physicians to make accurate and prompt diagnoses of a wide range of medical conditions [1]. The automated classification systems can assist in the activities such as the identification of the brain tumors in the MRI scans or the identification of pneumonia in the chest X-rays. Such systems are capable of minimizing the effort required to diagnose a patient, minimize human error and achieve improved patient outcomes. However, manual visualization of medical images is time-consuming, may be affected by the subjective judgement, and requires an in-depth knowledge of the topic. As a result, computer-aided diagnosis (CAD) systems have been developed to handle this process in an automated way.

Convolutional neural networks are widely used in medical image analysis today. They perform well across many diagnostic tasks [2]. Architectures such as ResNet-50 [2], DenseNet-121 [3], EfficientNet [4], MobileNet-V2 [5], and VGG19 [6] learn features at different levels from images, which helps in classification. At the same time, most of these models still work on spatial data, meaning raw pixel values. Because of that, some finer details can get missed. This is especially true for high-frequency patterns like edges, textures, and small shape differences that are often useful for identifying diseases [7, 8].

The frequency domain can be obtained using the Fast Fourier Transform (FFT). It gives another way to look at images by splitting them into magnitude and phase components. These components often bring out patterns that are not easy to see in the usual spatial

view [9, 10]. This is useful in medical imaging, where small differences in tissue appearance, lesion shape, and variations within the same class often show up as high-frequency patterns and help in telling diseases apart. For example, different types of brain tumors show different spectral patterns, and pneumonia in chest X-rays produces frequency patterns that are not seen in healthy lungs.

1.2 Research Gaps

Frequency-domain learning is promising, but a handful of practical gaps keep it from being a drop-in replacement for a purely spatial pipeline:

1. **Single-scale frequency analysis:** Most existing frequency-aware CNNs attach a spectral block at a single depth, which ignores the fact that pathological cues appear at several spatial scales and rarely at only one of them.
2. **Lack of explainability:** Spectral features are notoriously hard to narrate back to a clinician; a peak in the magnitude map does not localise cleanly to a region on the original scan, and this translation gap is a real barrier to adoption.
3. **Feature fusion challenges:** Spatial activations and frequency activations live on very different scales, and naively stacking them tends to produce numerically unstable training; simple concatenation or addition also fails to exploit the complementary information that different layers of the backbone carry.
4. **Phase sensitivity:** Phase carries most of the positional information in the spectrum, yet it is also highly discontinuous and noisy. Feeding the raw phase into a CNN almost always hurts performance unless the encoding is handled carefully.
5. **Cross-domain generalisability:** Published frequency-domain results are often tied to a single benchmark. The open question is whether the same spectral ideas carry over when the modality changes (MRI vs. radiography) and when the classification task itself changes (multi-class structured tumours vs. binary diffuse opacities).

1.3 Problem Statement

Addressing the loss of fine-grained high-frequency details due to repeated downsampling by proposing a multi-scale frequency pyramid fusion framework that integrates FFT-based representations from intermediate layers using learnable gating for enhanced discrimination and interpretability.

1.4 Objectives

We approach those gaps in two phases:

- **Phase 1:** Build a dual-frequency feature-fusion pipeline that lifts the spatial image into a 9-channel spectral tensor, merges the spatial and spectral streams through a learnable channel-attention block, and supports frequency-domain class activation mapping for interpretability.
- **Phase 2:** Extend that pipeline into a multi-scale frequency pyramid. Frequency-aware attention is applied at several depths of the backbone, a top-down feature pyramid knits the scales together, and a hierarchical, frequency-aware CAM is used to explain the decisions back to a clinical reader.

1.5 Contributions

Concretely, this thesis contributes:

1. A **9-channel spectral representation** built from the FFT, in which log-magnitude is paired with sine/cosine encodings of the phase so that both spectral energy and spatial positioning are preserved.
2. A **Dual-Frequency Feature Fusion (DFFF)** module that fuses spatial and spectral activations through channel attention with learnable domain weights.
3. A **Hierarchical Frequency-Spatial Attention (HFSA)** architecture in which frequency-attention gates operate at multiple depths and a feature pyramid network is combined with a learnable softmax-weighted fusion rule.

4. A **frequency-domain CAM** variant that maps spectral attention back into the spatial image, producing clinically interpretable saliency maps that emphasise disease-relevant regions.
5. A cross-modality evaluation conducted on two medical imaging datasets — the four-class **Brain Tumor MRI** benchmark and the binary **Chest X-ray Pneumonia** benchmark — across multiple CNN backbones, jointly assessing how effectively the approach generalises across different imaging modalities and task formulations.

Chapter 2

Related Work

Three strands of prior work bear directly on this thesis, and it is helpful to read them side by side: deep networks that deliberately move into the frequency domain, multi-scale feature pyramids in both natural and medical images, and the explainable-AI literature that grounds CAM-based interpretability for clinical CNNs. The rest of this chapter takes up each strand in turn.

2.1 Frequency-Domain Learning in CNNs

2.1.1 Fourier Transform in Deep Learning

Spectral operators have been working their way back into deep networks for some years now, largely because they capture structure that pure spatial convolutions tend to miss. Mathieu [11] showed early on that FFT-based convolutions could cut training wall-clock time substantially without any loss in accuracy, and Vasilache [12] carried that idea onto the GPU with the *fbfft* kernels, which still serve as a reference implementation for large-kernel spectral convolution.

Within medical imaging specifically, Nahid and Kong [13] paired DFT and DCT descriptors with a residual CNN — their CNN-DF and CNN-DC variants — and reported solid gains on histopathological breast-tissue classification, which is a useful early sign that spectral inputs are worth carrying through to a clinical task. Goh [8] then pushed in an efficiency direction with FDCNN, whose RFFT layers and channel-independent convolutions retain the frequency detail while trimming the compute.

Xu [7] took a step back and asked which parts of the spectrum a CNN actually relies on; using DCT features together with a dynamic channel-selection mechanism, they observed a marked bias toward low-frequency components and showed that choosing frequencies intelligently, rather than dumping the whole spectrum in, is what drives the gain. Totterstrom [10] draws an almost opposite lesson: raw Fourier spectra, fed directly into a CNN, actually underperform RGB inputs, which reinforces the need for carefully engineered spectral encodings rather than passing the DFT coefficients straight through.

2.1.2 Frequency-Domain Medical Image Analysis

On the segmentation side, Tang [14] folded FFT into a U-Net through a twin-kernel Fourier convolution (tKFC-Net), which kept global context and local edges in balance — a useful property for structures whose boundary is the clinically interesting feature. Sassali [9] surveyed the broader question of training deep networks on Fourier-transformed medical images and documented how different frequency bands contribute to downstream classification accuracy.

2.2 Multi-Scale Feature Pyramid Networks

2.2.1 Feature Pyramid Networks

Pathology does not settle on a single scale, which is exactly why multi-scale representation has been a recurring theme in visual recognition. The original Feature Pyramid Network of Lin et al. remains the canonical reference; its top-down pathway with lateral connections turns a single-scale backbone into a pyramid almost for free.

Van Quyen and Kim [15] showed that fusing predictions across several pyramid levels — rather than only at the top — yields real gains in real-time semantic segmentation, without the usual latency penalty that multi-scale inference tends to carry.

2.2.2 Multi-Scale Pyramid in Medical Imaging

Within the medical segmentation space, Zhang [16] built MS-Net around multi-scale attention modules (MSAM) stacked with feature pyramid modules (SFPM); the combination adapts the effective receptive field dynamically and pulls in context from a

wide range of depths. In a fusion setting, Liu [17] used residual-weighted pyramid paths to combine CT and MRI, producing cleaner fused outputs than single-scale baselines.

More recently, Xu [18] proposed MPCT, a hybrid that pairs pyramid convolutions with Transformer-style self-attention so that local structure and long-range dependencies can be modelled within the same network.

2.2.3 Frequency-Aware Multi-Scale Architectures

Xiao [19] sits squarely at the intersection of the previous two strands: MFN fuses multi-scale spatial features with frequency-domain descriptors for multi-class segmentation, and the authors argue convincingly that this combination sharpens boundary prediction when classes are heavily imbalanced.

Zhao [20]’s MADNet was designed for denoising rather than segmentation, but it is relevant here because its adaptive-learning units explicitly handle high- and low-frequency branches through an image-pyramid input — a concrete demonstration that frequency-aware processing really does pay off at multiple scales.

Zuo [21] reached a similar conclusion for infrared-visible image fusion: by decomposing both modalities into a pyramid and merging them level by level, their network recovered complementary information that single-scale methods discarded.

Beyond fusion, Zhang [22] used dense pyramidal convolutions together with multi-feature fusion for hyperspectral classification; their results show that hierarchical scale aggregation lets the network disambiguate spectrally similar classes that a single-scale baseline tends to conflate.

Chen [23]’s hybrid CNN-Transformer with multi-scale pyramid features fits the same pattern: the CNN side supplies the sharp local descriptors and the Transformer side carries long-range semantic context, and the pyramid is the bridge between them. CPFNet by Feng [24] completes the picture — its context pyramid fusion stages medical segmentation through a sequence of modules that each re-integrate information from the scale above.

2.3 Explainable AI for Medical Imaging

2.3.1 Limitations of LIME and SHAP

Explainable artificial intelligence (XAI) is the umbrella term for techniques that make the predictions of opaque models legible to a human reader [1]. Two of the most-cited model-agnostic approaches are LIME [25] and SHAP [26], and both carry well-known caveats when they are applied to medical images.

LIME works by fitting a simple, locally faithful surrogate to the black-box model around a particular input. For images the input is first partitioned into superpixels, and the surrogate is then explained over those regions. The catch is that superpixel-level explanations are coarse by construction: they do not give pixel-accurate localisation, and for deep models the perturbation-sampling step can take close to ten minutes per image [25].

SHAP, by contrast, grounds feature attribution in Shapley values from cooperative game theory, which gives it an appealing axiomatic backbone. In practice, however, a $224 \times 224 \times 3$ image has so many features that the exact Shapley computation demands 2^M subset evaluations and is therefore infeasible; the independence assumption built into most SHAP approximations also yields unrealistic pixel combinations that the original model was never trained on [26].

2.3.2 Class Activation Mapping Methods

CAM-style methods side-step much of that cost by living inside the CNN itself. Grad-CAM [27] uses the gradients of the target class with respect to the last convolutional feature map to highlight class-discriminative regions, and Grad-CAM++ [28] sharpens that localisation by weighting at the pixel level, which is particularly useful when multiple instances of the same class are present. XGrad-CAM [29] reformulates the gradient attribution so that it satisfies a small set of axioms and, in doing so, yields more faithful maps. Smooth Grad-CAM [30] tackles the noise problem directly by averaging activations under small input perturbations — the resulting maps are less jagged and the object localisation is cleaner. Eigen-CAM [31] bypasses both the class score and the backward pass and instead reads off the principal components of the learnt feature maps, which produces surprisingly competitive visualisations at a fraction of the cost.

Score-CAM [32] is the method we carry forward into this thesis, and it is worth being explicit about why. It assigns a weight to each activation channel based on the softmax score obtained by masking the input with the upsampled and normalised activation map. No gradients are used at any stage. This is useful because the resulting maps are less noisy and less saturated, and it matters particularly in our case, since gradients would otherwise have to pass through FFT operations, which can be numerically sensitive.

Chapter 3

Dual-Frequency Feature Fusion Framework

This chapter presents the baseline spectral learning pipeline used throughout the thesis.

3.1 System Overview

Phase 1 can be read as a sequence of four stages connected end to end. An RGB medical image is first mapped into the frequency domain and represented as a 9-channel spectral tensor. This tensor is then fed into a CNN whose input layer is modified to accept spectral data in place of the standard 3-channel RGB input. At the end of the backbone, a dual-frequency fusion block combines spatial and spectral features through a learnable channel attention mechanism. Finally, Score-CAM combined with a frequency-to-spatial IFFT mapping produces human-readable saliency, closing the loop with the clinical reader.

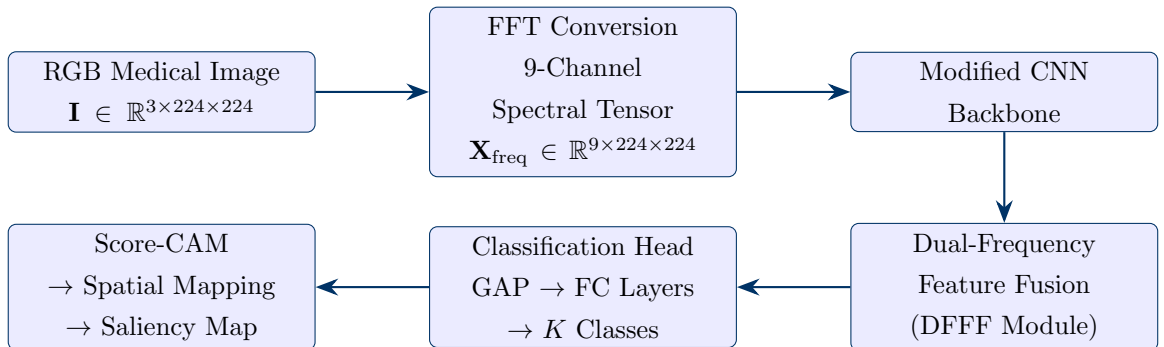


FIGURE 3.1: Dual-Frequency Feature Fusion framework architecture

3.2 Spatial-to-Frequency Domain Conversion

An RGB image, which has three color channels ($c \in \{R, G, B\}$) is converted into the frequency domain by using the 2D Discrete Fourier Transform on each channel individually:

$$F_c(u, v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f_c(x, y) \cdot e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})} \quad (3.1)$$

Here u, v are the spatial-frequency coordinates and M, N are the image dimensions. After centring with `fftshift`, we pull out the magnitude and phase components separately:

$$M_c(u, v) = |F_c(u, v)| = \sqrt{\text{Re}(F_c)^2 + \text{Im}(F_c)^2} \quad (3.2)$$

$$\Phi_c(u, v) = \arg(F_c(u, v)) = \arctan\left(\frac{\text{Im}(F_c)}{\text{Re}(F_c)}\right) \quad (3.3)$$

The magnitude map carries the spectral energy (and, with it, most of the texture and edge information), while the phase map keeps track of the spatial layout. The magnitudes span several orders of magnitude, so we compress them logarithmically before they reach the network:

$$L_c(u, v) = \ln(\max(M_c(u, v), \epsilon) + \epsilon), \quad \epsilon = 10^{-6} \quad (3.4)$$

Phase is 2π -periodic and discontinuous at the wrap-around, which makes it very hard for a CNN to learn from directly. We therefore encode it through a pair of trigonometric functions:

$$P_c^{\cos}(u, v) = \cos(\Phi_c(u, v)), \quad P_c^{\sin}(u, v) = \sin(\Phi_c(u, v)) \quad (3.5)$$

Stacking the three derived maps of each colour channel gives the 9-channel spectral tensor used downstream:

$$\mathbf{X}_{\text{freq}} = [L_R, P_R^{\cos}, P_R^{\sin}, L_G, P_G^{\cos}, P_G^{\sin}, L_B, P_B^{\cos}, P_B^{\sin}] \in \mathbb{R}^{9 \times H \times W} \quad (3.6)$$

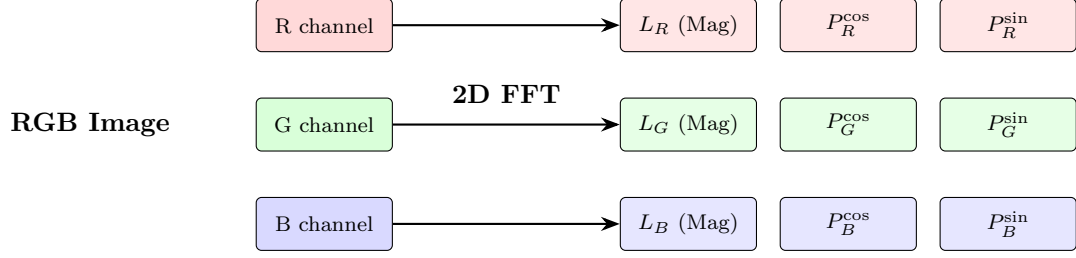


FIGURE 3.2: 9-channel spectral decomposition of an RGB image

3.3 Modified CNN Backbone

The backbone is a standard CNN with a single surgical change: its first convolution is rewritten to accept a 9-channel spectral input instead of the usual 3-channel RGB:

$$\text{Conv}_1 : \mathbb{R}^{9 \times H \times W} \rightarrow \mathbb{R}^{64 \times \frac{H}{2} \times \frac{W}{2}} \quad (3.7)$$

To avoid throwing away everything the pre-trained backbone already knows, we initialise this widened layer with a transfer-and-scale rule:

$$\mathbf{W}_{\text{conv1}}[:, 0:3, :, :] = \mathbf{W}_{\text{pretrained}} \quad (3.8)$$

$$\mathbf{W}_{\text{conv1}}[:, 3:9, :, :] = 0.1 \cdot \mathbf{W}_{\text{Kaiming}} \quad (3.9)$$

The factor of 0.1 keeps the six new frequency channels quiet early in training, so that they refine rather than overrun the spatial features the backbone was pre-trained on. After the stem, the tensor flows through the usual four residual stages of the backbone, each of which implements:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x} \quad (3.10)$$

where $\mathcal{F}$ is the usual Conv-BN-ReLU residual block. The final stage produces deep feature maps $\mathbf{A} \in \mathbb{R}^{B \times C_{\text{backbone}} \times h \times w}$, which are the input to the fusion module that follows.

3.4 Dual-Frequency Feature Fusion (DFFF) Module

The DFFF module operates on the activations $\mathbf{A} \in \mathbb{R}^{B \times C \times h \times w}$ produced by the last residual block. A second 2D FFT is taken on these activations to expose their spectral content:

$$G_k(u, v) = \text{FFT}_{2D}(A_k(x, y)), \quad M_k^{\text{feat}} = |G_k| \quad (3.11)$$

The magnitude spectra are log-scaled and normalised to obtain $\mathbf{A}_{\text{freq}}$. Because the two streams are on very different numerical scales, each is fed through its own BatchNorm layer:

$$\tilde{\mathbf{A}} = \text{BN}_{\text{spatial}}(\mathbf{A}), \quad \tilde{\mathbf{A}}_{\text{freq}} = \text{BN}_{\text{freq}}(\mathbf{A}_{\text{freq}}) \quad (3.12)$$

The two normalised streams are concatenated along the channel axis and passed through a squeeze-excitation-style channel attention block:

$$\mathbf{C} = [\tilde{\mathbf{A}}; \tilde{\mathbf{A}}_{\text{freq}}] \in \mathbb{R}^{B \times 2C \times h \times w} \quad (3.13)$$

$$\mathbf{z} = \text{GAP}(\mathbf{C}), \quad \mathbf{s} = \sigma(\mathbf{W}_2 \cdot \delta(\mathbf{W}_1 \cdot \mathbf{z})) \quad (3.14)$$

$$\hat{\mathbf{C}} = \mathbf{s} \odot \mathbf{C} \quad (3.15)$$

Here $\mathbf{W}_1 \in \mathbb{R}^{C \times 2C}$ squeezes the channel dimension and $\mathbf{W}_2 \in \mathbb{R}^{2C \times C}$ restores it. The attention-weighted tensor is split back into its spatial and frequency halves and re-combined using learnable domain weights:

$$\mathbf{A}_{\text{fused}} = \alpha \cdot \hat{\mathbf{A}}_s + \beta \cdot \hat{\mathbf{A}}_f \quad (3.16)$$

The weights $\alpha = \sigma(\alpha_{\text{raw}})$ and $\beta = \sigma(\beta_{\text{raw}})$ are learnt end-to-end and initialised asymmetrically to $\alpha_{\text{init}} = 0.7$ and $\beta_{\text{init}} = 0.3$, so that the spatial stream dominates at the

start of training and the spectral contribution is ramped up only once the frequency channels are themselves well-calibrated.

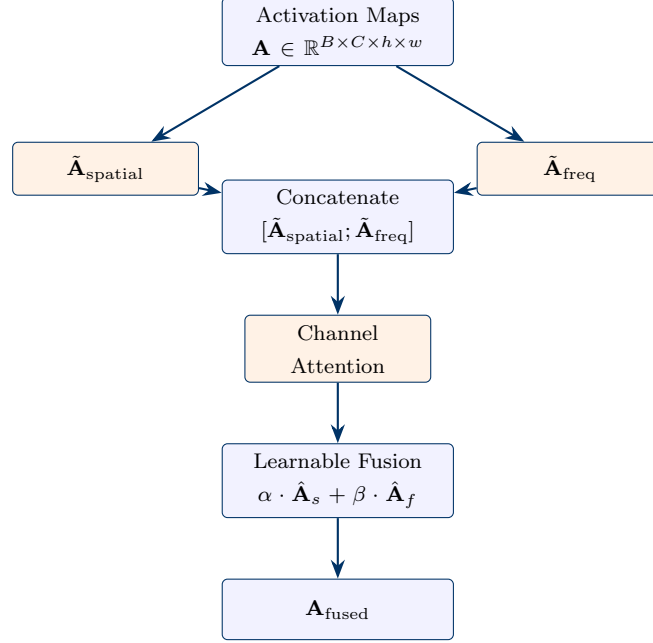


FIGURE 3.3: Architecture of the DFFF module

3.5 Classification Head

The fused feature map is globally pooled and passed through a three-layer fully connected head:

$$\mathbf{v} = \frac{1}{h \cdot w} \sum_{i=1}^h \sum_{j=1}^w \mathbf{A}_{\text{fused}}(:, :, i, j) \in \mathbb{R}^{B \times C_{\text{backbone}}} \quad (3.17)$$

$$\mathbf{v} \xrightarrow{\text{Drop}(p_1)} \xrightarrow{W_1} \xrightarrow{\text{ReLU}} \xrightarrow{\text{BN}} \xrightarrow{\text{Drop}(p_2)} \xrightarrow{W_2} \xrightarrow{\text{ReLU}} \xrightarrow{\text{BN}} \xrightarrow{\text{Drop}(p_3)} \xrightarrow{W_3} \mathbf{o} \quad (3.18)$$

The output $\mathbf{o} \in \mathbb{R}^{B \times K}$ holds the K class logits. For brain tumour classification $K = 4$ (glioma, meningioma, pituitary, no tumour); for chest X-ray pneumonia detection $K = 2$ (normal, pneumonia). The dropout schedule is deliberately top-heavy — $p_1 > p_2 > p_3$ — so that regularisation is strongest on the widest, most over-parameterised layers and tapers off as the feature dimension contracts.

3.6 Explainability

Score-CAM [32] operates on the fused activations $\mathbf{A}_{\text{fused}} \in \mathbb{R}^{1 \times C \times h \times w}$. Each channel is first upsampled back to the input resolution and then min-max normalised to $[0, 1]$:

$$\bar{A}_k = \frac{A_k^\uparrow - \min(A_k^\uparrow)}{\max(A_k^\uparrow) - \min(A_k^\uparrow)} \quad (3.19)$$

The normalised map $\bar{A}_k$ is then used as a soft mask on the 9-channel spectral input: $X_k^* = X_{\text{freq}} \odot \bar{A}_k$. Running the masked input through the frozen model yields the confidence of the target class t :

$$s_k = \text{softmax}(f(X_k^*))_t \quad (3.20)$$

The final CAM is the ReLU of the confidence-weighted sum of the original (un-masked) activations:

$$\text{CAM}(u, v) = \text{ReLU} \left(\sum_{k=1}^C s_k \cdot A_k(u, v) \right) \quad (3.21)$$

3.7 Frequency-to-Spatial Domain Mapping

The CAM produced above still lives in the frequency domain. To make it legible at the bedside we map it back into pixel space via the inverse FFT of the masked spectrum:

$$I_{\text{sal}} = \text{IFFT}_{2\text{D}} (\text{CAM} \cdot M \cdot e^{j\Phi}) \quad (3.22)$$

A short post-processing chain then smooths and sharpens the saliency response. First, a Gaussian filter with $\sigma = 2.5$ removes high-frequency ringing introduced by the IFFT; second, a 40th-percentile threshold suppresses background noise; third, a 5×5 morphological closing followed by a 3×3 opening cleans up the foreground masks; and, last, a $\gamma = 0.7$ power-law transformation boosts the contrast of the remaining response. The processed map is then blended back onto the original scan, which gives the clinician an overlay that points directly at the regions driving the prediction — tumour tissue in MRI, opacified lung fields in chest X-ray.

Chapter 4

Multi-Scale Frequency Pyramid Fusion

This chapter builds on the DFFF pipeline of the previous chapter and extends it into the Multi-Scale Frequency Pyramid Fusion (MSPF) architecture, whose central mechanism is Hierarchical Frequency-Spatial Attention (HFSA).

4.1 Motivation for Multi-Scale Extension

DFFF inserts its spectral fusion block at a single depth — the last residual stage of the backbone, whose feature maps measure 7×7 . That choice works, but in hindsight it leaves several useful signals on the table:

- The higher-resolution feature maps earlier in the network (Layer 2 at 28×28 and Layer 3 at 14×14) still carry the fine-grained texture and edge information that is most diagnostic when a lesion boundary must be delineated; a single 7×7 injection point throws most of that away.
- Conversely, 7×7 is too coarse to support high-frequency spectral analysis on its own; once the receptive field has grown this much, the FFT is effectively looking at a small number of samples and cannot resolve fine spectral detail.
- Real pathology does not present at one scale. Global shape cues, internal heterogeneity and fine edge detail coexist in almost every clinical image, and a

single-depth fusion module inherently mixes these into one stream rather than letting each scale be processed on its own terms.

MSPF addresses these points by making frequency-aware processing active at several depths of the backbone simultaneously, and then gluing those depths together with a feature pyramid network that learns how to weight them.

4.2 Hierarchical Frequency-Spatial Attention (HFSA)

HFSA is made up of three cooperating pieces that attach at different depths of the modified backbone:

9-ch Frequency Input

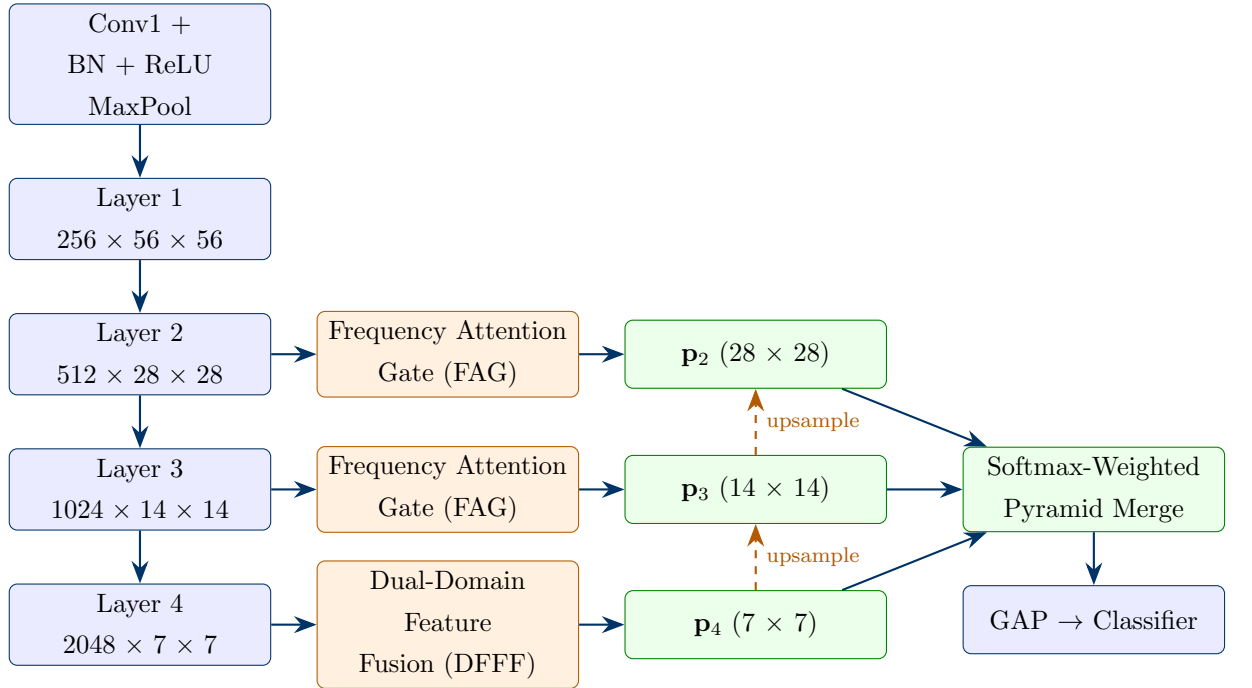


FIGURE 4.1: Architecture of the Multi-Scale Frequency Pyramid Fusion (MSPF) framework with HFSA

4.2.1 Frequency Attention Gate (FAG)

FAG is inserted at Layer 2 (28×28) and Layer 3 (14×14), where the activations still retain enough spatial resolution for a useful FFT. Given activation maps $\mathbf{x} \in \mathbb{R}^{B \times C \times h \times w}$, the gate proceeds in three steps.

Step 1: Compute FFT magnitude of activation maps:

$$\mathbf{x}_{\text{freq}} = |\text{FFT}_{2\text{D}}(\mathbf{x})| \quad (4.1)$$

Step 2: Process through a lightweight convolutional gate network:

$$\mathbf{g} = \sigma(\text{Conv}_{1 \times 1}(\delta(\text{BN}(\text{Conv}_{3 \times 3}(\mathbf{x}_{\text{freq}})))) \quad (4.2)$$

The 3×3 convolution squeezes the channel dimension by the reduction ratio r and the 1×1 convolution re-expands it back to C , which keeps the gate lightweight in both parameters and compute.

Step 3: Apply residual-style gating:

$$\mathbf{x}_{\text{out}} = \mathbf{x} \odot (1 + \alpha \cdot \mathbf{g}) \quad (4.3)$$

where α is a learnable scalar initialised to 0.1. The residual form is deliberate: the identity term (1) guarantees that the spatial signal is never overwritten, while the gated frequency term ($\alpha \cdot \mathbf{g}$) supplies only a refinement. Without this safety net the two streams — which are on different numerical scales — tend to interact badly and destabilise training in the early epochs.

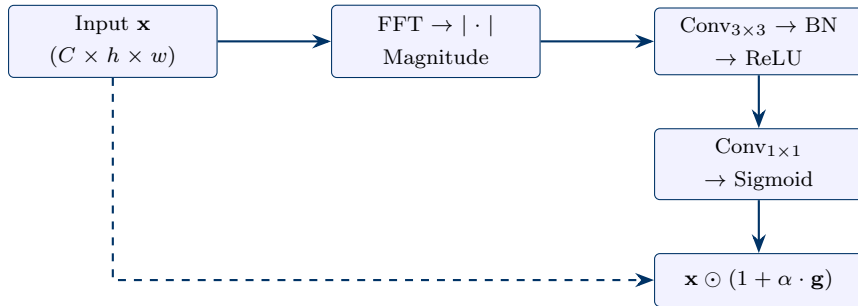


FIGURE 4.2: Architecture of Frequency Attention Gate (FAG)

4.2.2 Dual-Frequency Feature Fusion at Layer 4

The full DFFF block from Phase 1 is reused at Layer 4 (7×7), where it performs complete spatial-frequency fusion on the most abstract features in the network. The asymmetry between Layers 2–3 and Layer 4 is intentional:

- At the higher resolutions of Layer 2 and Layer 3, the lightweight FAG gate is enough; inserting a full DFFF block there would blow up the compute budget and re-introduce the instability that FAG is designed to avoid.
- At Layer 4, on the other hand, the spatial dimensions are small enough that a second FFT is essentially free, and the features are abstract enough that a richer cross-domain fusion is worth the extra parameters.

4.3 Feature Pyramid Network (FPN) with Learnable Fusion

Once each scale has been processed by its own frequency-aware block, the three feature maps are glued together with a top-down FPN.

4.3.1 Top-Down Pathway

P4 construction. The DFFF output at Layer 4 acts as the coarsest rung of the pyramid:

$$\mathbf{p}_4 = \text{DFFF}(\mathbf{x}_4) \in \mathbb{R}^{B \times 2048 \times 7 \times 7} \quad (4.4)$$

P3 construction. $\mathbf{p}_4$ is upsampled to 14×14 and added to the laterally projected Layer 3 output:

$$\mathbf{p}_3 = \text{Upsample}(\mathbf{p}_4, 14 \times 14) + \text{Lat}_3(\text{FAG}_3(\mathbf{x}_3)) \quad (4.5)$$

where Lat_3 is a 1×1 convolution that lifts the Layer-3 channel count from 1024 up to 2048.

P2 construction. The same recipe is repeated one step higher up the pyramid:

$$\mathbf{p}_2 = \text{Upsample}(\mathbf{p}_3, 28 \times 28) + \text{Lat}_2(\text{FAG}_2(\mathbf{x}_2)) \quad (4.6)$$

4.3.2 Adaptive Pyramid Fusion

All three pyramid levels are first brought to a common resolution of 7×7 , and then merged with a set of softmax-normalised weights that are themselves learnt end-to-end:

$$\mathbf{p}_2^s = \text{AdaptiveAvgPool}(\mathbf{p}_2, 7 \times 7), \quad \mathbf{p}_3^s = \text{AdaptiveAvgPool}(\mathbf{p}_3, 7 \times 7), \quad \mathbf{p}_4^s = \mathbf{p}_4 \quad (4.7)$$

$$\mathbf{w} = \text{softmax}(\boldsymbol{\ell}), \quad \boldsymbol{\ell} \in \mathbb{R}^3 \text{ (learnable logits)} \quad (4.8)$$

$$\mathbf{M}_{\text{merged}} = w_1 \cdot \mathbf{p}_2^s + w_2 \cdot \mathbf{p}_3^s + w_3 \cdot \mathbf{p}_4^s \quad (4.9)$$

A final 3×3 refinement convolution, followed by BatchNorm and ReLU, produces the output used by the classifier:

$$\mathbf{M}_{\text{out}} = \text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3}(\mathbf{M}_{\text{merged}}))) \quad (4.10)$$

Constraining the weights to a probability simplex via softmax is what lets the model decide, per dataset, how much of the output should come from each pyramid level. The right balance varies with the pathology: coarse semantic features are sufficient when lesions are large and well-defined, but the finer, higher-resolution pyramid rungs become critical for diffuse or low-contrast findings.

4.4 Multi-Scale Explainability

Score-CAM is lifted to the multi-scale setting by running it on the fused HFPN output rather than on a single layer of the backbone:

1. Take the fused activations at the top of the HFPN: $\mathbf{A}_{\text{fused}} \in \mathbb{R}^{1 \times 2048 \times 7 \times 7}$.
2. Upsample each channel to the input resolution of 224×224 and min-max normalise to $[0, 1]$.
3. Treat each normalised channel as a soft mask over the 9-channel spectral input.
4. Run each masked input through the full model and read off the softmax score of the target class.
5. Combine the original activations, weighted by those per-channel confidence scores, using Equation 3.21.

Because the HFPN output is itself an aggregate of all three pyramid levels, the resulting CAM inherits spectral information from Layer 2, Layer 3 and Layer 4 at once, and it is noticeably cleaner than the single-scale variant from Chapter 1’s baseline. The frequency-to-spatial mapping described in Section 3.7 turns it into a saliency overlay that a clinician can read directly on the original scan.

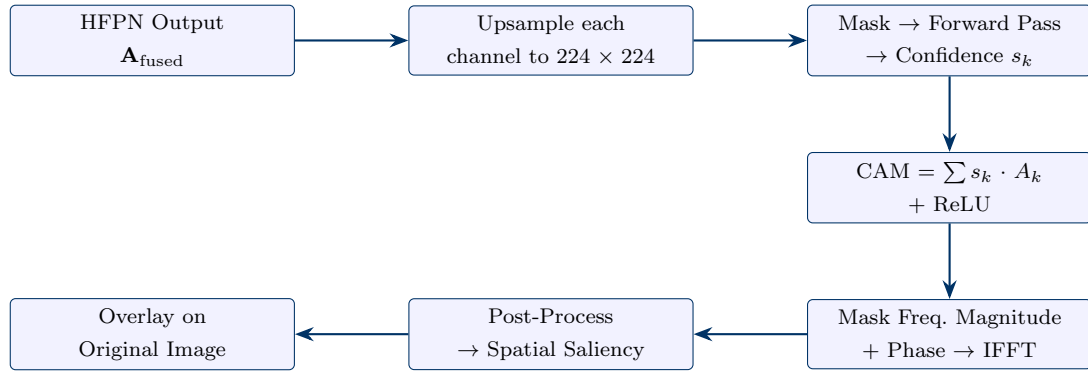


FIGURE 4.3: Multi-scale Score-CAM pipeline

Chapter 5

Experimental Setup

5.1 Datasets

To probe the generality of the proposed framework across both imaging modality and task structure, we evaluate on two publicly available benchmarks that differ on almost every axis that matters clinically:

5.1.1 Brain Tumor MRI Dataset

The [Brain Tumor MRI dataset](#) distributed on Kaggle by Masoud Nickparvar contains T1-weighted contrast-enhanced MRI volumes for multi-class brain tumour classification, spread across four categories: glioma, meningioma, pituitary and no tumour.

TABLE 5.1: Brain Tumor MRI dataset statistics.

Class	Training Set	Testing Set	Total
Glioma	1,321	400	1,721
Meningioma	1,339	400	1,739
No Tumor	1,595	400	1,995
Pituitary	1,457	400	1,857
Total	5,712	1,600	7,312

From the training portion we carve out an 85%/15% train/validation split with a fixed seed of 42, which yields roughly 4,855 training and 857 validation samples.

5.1.2 Chest X-Ray Pneumonia Dataset

The second dataset is the [Chest X-Ray Pneumonia dataset](#) hosted on Kaggle by Paul Mooney, which itself comes from the Kermany et al. release. It contains antero-posterior chest X-rays and is organised into two classes — **Normal** and **Pneumonia** — with the pneumonia class lumping together bacterial and viral cases.

TABLE 5.2: Chest X-Ray Pneumonia dataset statistics.

Class	Training Set	Testing Set	Total
Normal	1,341	234	1,575
Pneumonia	3,875	390	4,265
Total	5,216	624	5,840

Taken together, the two datasets stress the framework along complementary axes: a different imaging modality (radiography rather than MRI), a different label structure (an imbalanced binary task instead of a near-balanced four-way classification), and a different kind of pathology (diffuse pulmonary opacities rather than focal mass lesions). The same 85%/15% train/validation split is used here as well.

5.2 Data Preprocessing

5.2.1 Test Set Preprocessing

The test pipeline is deliberately deterministic: every image is resized to 224×224 , converted to a tensor, normalised with the standard ImageNet statistics and then lifted into the 9-channel spectral tensor via FFT. No augmentation is used at test time.

5.2.2 Training Set Augmentation

On the training side, stochastic augmentation is layered on top of the same backbone preprocessing to improve generalisation:

TABLE 5.3: Training data augmentation pipeline.

Augmentation	Parameters
Resize	224×224
Random Horizontal Flip	Probability = 0.5
Random Vertical Flip	Probability = 0.3
Random Rotation	± 20
Color Jitter	Brightness = ± 0.2 , Contrast = ± 0.2 , Saturation = ± 0.1
Random Affine	Translation $\leq 10\%$, Scale $\in [0.9, 1.1]$
Normalize	Mean = $[0.485, 0.456, 0.406]$, Std = $[0.229, 0.224, 0.225]$ (ImageNet)
FFT Conversion	9-channel spectral tensor (Section 3.2)

5.3 Training Strategy

5.3.1 Loss Function

The model is trained with cross-entropy loss augmented by label smoothing ($\epsilon = 0.1$):

$$\mathcal{L} = - \sum_{k=1}^K \tilde{y}_k \log(\hat{y}_k), \quad \tilde{y}_k = (1 - \epsilon)y_k + \frac{\epsilon}{K} \quad (5.1)$$

with $\hat{y} = \text{softmax}(\mathbf{o})$. Label smoothing discourages the softmax from becoming overconfident on any single class and in practice trades a small amount of training accuracy for a meaningfully better validation curve.

5.3.2 Optimizer and Learning Rate

We optimise with AdamW using decoupled weight decay $\lambda = 5 \times 10^{-4}$, and apply differential learning rates so that the pre-trained backbone is updated more gently than the newly introduced modules:

- Pre-trained backbone layers: $\eta_{\text{pre}} = \eta \times 0.01$
- New modules (HFPN, FAG, classifier): $\eta_{\text{new}} = \eta \times 0.5$

The learning rate itself follows a cosine-annealing schedule with warm restarts:

$$\eta(t) = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos \left(\frac{T_{\text{cur}}}{T_i} \pi \right) \right) \quad (5.2)$$

with an initial restart period of $T_0 = 15$ epochs, a period multiplier of $T_{\text{mult}} = 2$ and a floor of $\eta_{\min} = 10^{-7}$.

5.3.3 Training Stability

TABLE 5.4: Training stability mechanisms.

Mechanism	Configuration
Mixed Precision (AMP)	float16 forward, float32 weight updates
Gradient Clipping	Global norm clipping, max norm = 1.0
NaN/Inf Safeguards	<code>torch.nan_to_num</code> at critical points
Early Stopping	Patience = 15 epochs

5.4 Hardware and Software

The codebase is written in PyTorch and draws on torchvision for the pre-trained backbone weights and the built-in augmentation primitives. Training, validation and inference were all run on a single NVIDIA GPU with 16 GB of video memory, which proved sufficient to hold a batch of 9-channel spectral inputs together with the full multi-scale feature hierarchy in memory. We enable `cudnn.benchmark` so that cuDNN can autotune the fastest kernel for each convolution shape, and intentionally leave deterministic mode off for the speed gain it brings. Automatic Mixed Precision (AMP) together with `GradScaler` handles forward passes in float16 while keeping the gradients and weight updates in float32; the result is a noticeable throughput improvement with no measurable loss of accuracy. Post-processing of saliency maps is done with SciPy, and all of the plots in this report were produced with Matplotlib.

5.5 Hyperparameters

TABLE 5.5: Hyperparameter configuration for MSPF.

Hyperparameter	Value
Epochs (max)	100
Batch Size	64
Base Learning Rate	0.001
Pre-trained LR Scale	$\times 0.01$
New Modules LR Scale	$\times 0.5$
Weight Decay	5×10^{-4}
Dropout Rate	0.5 (progressive: $p_1 = 0.5, p_2 = 0.35, p_3 = 0.25$)
Label Smoothing	$\epsilon = 0.1$
Gradient Clip Norm	1.0
FAG Reduction Ratio	8
FAG α Init	0.1
DFFF α / β Init	0.7 / 0.3
Cosine Schedule T_0	15
Cosine Schedule T_{mult}	2
Early Stop Patience	15

Chapter 6

Results and Analysis

This chapter reports both the quantitative and the qualitative findings obtained with the Multi-Scale Frequency Pyramid Fusion architecture on our two benchmarks — the four-class Brain Tumor MRI dataset and the binary Chest X-Ray Pneumonia dataset.

6.1 Brain Tumor MRI Dataset Results

6.1.1 Multi-Backbone Evaluation

TABLE 6.1: Comparative performance of CNN backbones with the proposed MSPF framework on the brain tumor MRI test set.

Metric	ResNet-50	MobileNetV2	DenseNet-121	EfficientNet-B0	VGG-19
Overall Accuracy (%)	82.75	79.44	85.63	83.88	78.19
Cohen’s Kappa Score (%)	77.00	72.59	80.84	78.51	70.92
Overall Precision (%)	83.26	79.87	86.11	84.22	78.54
Overall Recall (%)	82.75	79.44	85.63	83.88	78.19
Overall F1 Score (%)	82.29	79.02	85.38	83.62	77.73
Overall Specificity (%)	94.25	93.15	95.21	94.63	92.73

6.1.2 Training Dynamics

Training of the ResNet-50 variant of MSPF on the Brain Tumor MRI dataset ran for 59 epochs before the early-stopping criterion fired. A few observations from that run are worth highlighting:

- The best validation accuracy — **87.26%** — was reached at epoch 44.
- The validation curve converges in the punctuated style that cosine annealing with warm restarts tends to produce; each restart temporarily lifts the learning rate and pulls the optimiser out of the plateau it was sitting in, and the model is visibly better after each such kick.
- Training was terminated at epoch 59 once 15 consecutive epochs had passed without a new best validation score.

6.1.3 Test Set Performance

TABLE 6.2: MSPF classification report on brain tumor MRI test set.

Class	Precision (%)	Recall (%)	F1 Score (%)	Support
Glioma	87	62	72	400
Meningioma	71	76	73	400
No Tumor	82	99	90	400
Pituitary	93	94	93	400
Macro Avg	83	83	82	1600
Weighted Avg	83	83	82	1600
Overall Test Accuracy				82.75%

6.1.4 Per-Class Accuracy Analysis

TABLE 6.3: Per-class test accuracy for MSPF on brain tumor MRI.

Class	Correct / Total	Accuracy (%)
Glioma	249 / 400	62.25
Meningioma	305 / 400	76.25
No Tumor	395 / 400	98.75
Pituitary	375 / 400	93.75
Overall	1,324 / 1,600	82.75

6.1.5 Analysis

Strong performance on the No Tumor and Pituitary classes. Recall on the No-Tumor class reaches 99%, which is the behaviour we want from a screening model: it very rarely misses a healthy brain. Pituitary tumours, whose shape cues are typically clean and well-localised, are recovered at 93.75% accuracy.

Glioma is the hard class. Glioma has the lowest per-class accuracy at 62.25%. This is not surprising, since gliomas vary considerably in structure. Diffuse gliomas often have unclear boundaries and varied internal patterns, which can resemble meningiomas. The precision and recall values reflect this clearly. The model reaches a high precision of 87% but a lower recall of 62%, meaning that when it predicts glioma, it is usually correct, but some actual glioma cases are being classified as meningiomas.

The multi-scale pyramid is pulling its weight. Because the FPN gives the classifier access to the fine 28×28 textures from Layer 2 alongside the coarser 7×7 semantics from Layer 4, and because the learnable softmax weights decide how much of each to use, the model ends up with a genuinely hybrid view of each sample — coarse enough to separate tumour from non-tumour, fine enough to keep the subtype boundaries.

Score-CAM sample predictions. On representative test slices the model is decisive without being over-confident: 77.2% for meningioma, up to 89.8% for glioma, and 94.5% for no tumor. More importantly, the saliency maps put their mass on the correct anatomy, the tumour region where one is present, and the surrounding parenchyma where one is not.

6.2 Chest X-Ray Pneumonia Dataset Results

6.2.1 Multi-Backbone Evaluation

TABLE 6.4: Comparative performance of CNN backbones with the proposed MSPF framework on the Chest X-Ray Pneumonia test set.

Metric	ResNet-50	MobileNetV2	DenseNet-121	EfficientNet-B0	VGG-19
Overall Accuracy (%)	81.41	79.17	84.45	82.37	77.56
Cohen’s Kappa Score (%)	57.66	52.30	65.01	60.00	48.44
Overall Precision (%)	83.20	81.14	85.80	84.07	80.23
Overall Recall (%)	77.01	74.36	80.81	78.20	72.48
Overall F1 Score (%)	78.49	75.88	82.16	79.66	74.23
Overall Specificity (%)	77.01	74.36	80.81	78.20	72.48

6.2.2 Training Dynamics

On the Chest X-Ray Pneumonia dataset, the ResNet-50 variant of MSPF trained for 43 epochs before early stopping kicked in:

- Peak validation accuracy of **95.34%** was achieved at epoch 28.
- Convergence was noticeably faster than on the brain-tumour task; cosine annealing with warm restarts again drove most of the late-stage gains, and training accuracy topped out at 95.93% before the run was stopped.
- Early stopping terminated the run at epoch 43 after 15 epochs without improvement.
- The strong class imbalance was handled with a `WeightedRandomSampler` feeding a class-weighted cross-entropy loss, together with a lighter label smoothing of $\epsilon = 0.05$ (tuned down from the multi-class setting).

6.2.3 Test Set Performance

For the cross-modality evaluation on chest X-ray pneumonia ($K = 2$), the entire backbone, DFFF block and HFSA stack are reused unchanged; the only modification is the final classifier, whose output dimension is retargeted from four to two. The preprocessing pipeline — including the 9-channel spectral lift — is shared with the brain-tumour setup.

TABLE 6.5: MSPF classification report on the chest X-ray pneumonia test set.

Class	Precision (%)	Recall (%)	F1 Score (%)	Support
Normal	86.88	59.40	70.56	234
Pneumonia	79.53	94.62	86.42	390
Macro Avg	83.20	77.01	78.49	624
Weighted Avg	82.28	81.41	80.47	624
Overall Test Accuracy				81.41%

6.2.4 Per-Class Accuracy Analysis

TABLE 6.6: Per-class test accuracy for MSPF on Chest X-Ray Pneumonia.

Class	Correct / Total	Accuracy (%)
Normal	139 / 234	59.40
Pneumonia	369 / 390	94.62
Overall	508 / 624	81.41

6.2.5 Analysis

Pneumonia recall is the standout metric. The model identifies 94.62% of true pneumonia cases. This matters in a screening setting, where missing a diseased case is far more serious than incorrectly flagging a healthy one.

Normal-class recall is lower. Recall for the normal class is 59.40%. The model ends up predicting pneumonia more often, and this mostly comes down to the data distribution. Pneumonia cases are about 2.9 times more frequent than normal ones. Even with a `WeightedRandomSampler`, this imbalance does not fully disappear and still affects the decision boundary. Precision for the normal class is 86.88%, so when the model predicts *normal* it is usually correct. The issue is on the other side: many normal cases are still getting classified as pneumonia.

Cross-modality generalisation. The pneumonia experiment is used to see how the model behaves on a different type of data. MRI images are replaced with chest X-rays,

and the task itself shifts, from clear lesions to more diffuse opacities. This introduces a few challenges:

- **Class imbalance.** The dataset has a pneumonia-to-normal ratio of roughly 2.9:1. Pneumonia cases appear more often, which can bias the model. At the same time, the model still needs to detect pneumonia reliably without simply riding this imbalance.
- **Diffuse patterns.** Pneumonia does not present as a single clear region. It shows up as diffuse opacities spread across the lungs. The model therefore has to pick up signals that are distributed across the image, rather than concentrated in one place.
- **Different imaging type.** Chest X-rays are 2D projections of a 3D structure, and their contrast and noise profile differ from MRI. Even with this change, the same method still works, which shows that the approach is not tied to one imaging type.

Score-CAM sample predictions. For typical X-ray images, the softmax scores sit above 90% in most cases. Pneumonia examples fall around 91.6%, 91.8%, and 92.1%, while normal cases land around 96.2% and 96.3%. The saliency maps also look reasonable: on pneumonia scans, the model focuses on the affected lung regions; on normal scans, the response is more spread out across clear lung areas and does not latch onto unrelated structures.

6.3 Qualitative Analysis

6.3.1 Loss and Accuracy Graphs

To see the training behaviour of the MSPF model at a glance, we plot the loss and accuracy curves on both benchmarks; the two figures make it possible to read off how quickly the model was learning, how the cosine annealing schedule interacted with the loss landscape, and where the generalisation gap began to open up.

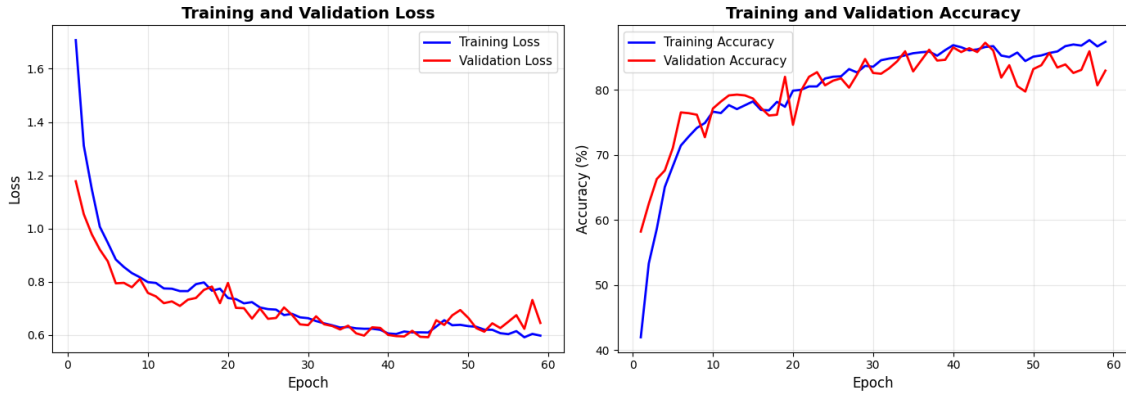


FIGURE 6.1: Training and validation loss and accuracy graphs for the Brain Tumor MRI dataset.

On the **Brain Tumor MRI** run (Figure 6.1) the training accuracy rises monotonically, while the validation accuracy is noticeably noisier: it climbs in jumps and then flattens out for extended stretches in between. Those discontinuities line up with the warm restarts of the cosine annealing schedule, which periodically lift the learning rate and let the optimiser escape from shallow local minima. Training and validation loss track each other closely through the first couple of restarts and then start to diverge around epoch 44, which is also where the best validation accuracy of 87.26% is logged. Beyond that point, the training loss continues to fall while the validation loss stalls — a mild but unmistakable sign of overfitting — and the early-stopping criterion fires at epoch 59.

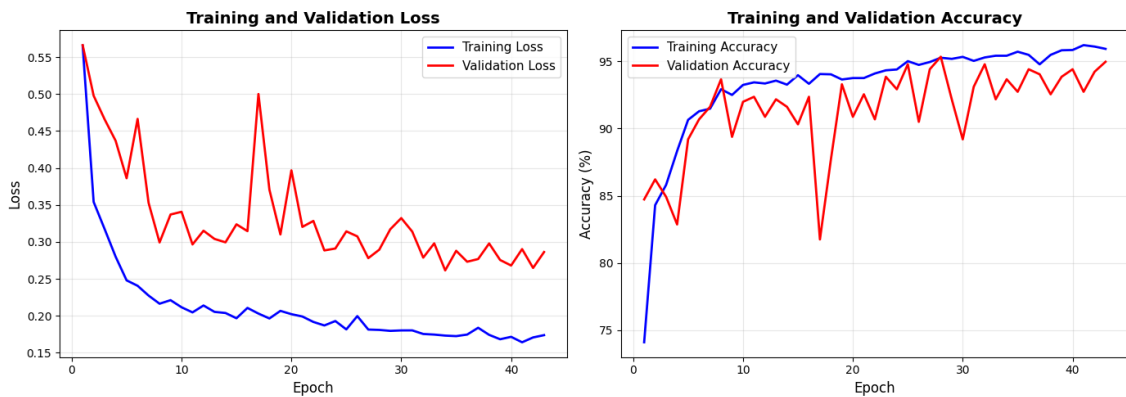


FIGURE 6.2: Training and validation loss and accuracy graphs for the Chest X-Ray Pneumonia dataset.

The **Chest X-Ray Pneumonia** run (Figure 6.2) tells a different story. The binary structure of the task makes early training much faster, and the model reaches its peak validation accuracy of 95.34% as early as epoch 28. Validation loss, however, swings

more violently than it does on the brain-tumour task — a direct consequence of the class imbalance and the diffuse, scattered nature of pneumonia opacities — and it is only the combination of `WeightedRandomSampler` and class-weighted cross-entropy that keeps the model from collapsing onto the majority class.

6.3.2 CAM Visualisations

The CAM family has grown substantially in recent years and now includes Grad-CAM [27], XGrad-CAM [29], Smooth Grad-CAM [30], Score-CAM [32], Ablation-CAM [33], Eigen-CAM [31] and Grad-CAM++ [28], each of which trades off fidelity, compute and noise differently. Within our pipeline we settle on Score-CAM: its outputs are spatially consistent, its attention concentrates cleanly on the anatomically relevant regions on both datasets, and the resulting heatmaps — both on brain MRI and on chest radiographs — are sharp enough to be read directly by a clinician.

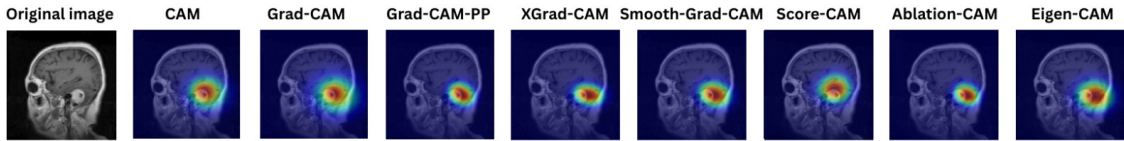


FIGURE 6.3: CAM variants visualisation for Brain Tumor MRI

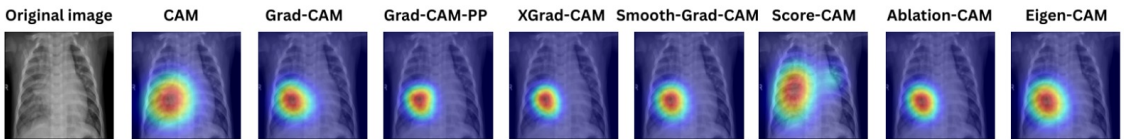


FIGURE 6.4: CAM variants visualisation for Chest X-ray Pneumonia

Compared with popular alternatives such as Grad-CAM and Grad-CAM++, Score-CAM delivered qualitatively better attributions in our setting, and several factors combine to explain that gap:

1. **Gradient independence.** Gradient-based methods like Grad-CAM are vulnerable to noisy or saturated gradients, and these problems are amplified inside a deep architecture like our multi-scale pyramid. Score-CAM dispenses with the backward pass entirely — it weights each activation channel by the softmax score obtained

from its masked forward pass — which removes a whole class of spurious positive attributions.

2. **Multi-scale frequency integration.** Because Score-CAM works with the fused activations at the top of the HFPN, it naturally mixes spatial information with the rescaled frequency cues from each pyramid level; we found that gradient-based CAM variants simply did not cope as well with the merged multi-scale representation.
3. **Sharper lesion boundaries.** By combining the coarse semantic outline with the high-resolution frequency detail, the resulting maps trace the outer edge of a lesion crisply instead of bleeding into the surrounding parenchyma. On brain MRI this shows up as a clean tumour boundary; on chest X-rays the lung opacities are highlighted without the response leaking into healthy lung tissue.
4. **Clinical interpretability and spectral specificity.** Standard spatial-domain CAMs tend to produce soft, blurred heatmaps, whereas the frequency-aware Score-CAM variant used here — combined with the IFFT-based mapping back to pixel space — converts spectral attention into sharply defined regions that a clinician can point to directly on the scan.

Chapter 7

Conclusion

7.1 Summary of Contributions

Taken end to end, this project lays out a two-stage approach to medical image classification in which frequency-domain learning and clinical interpretability are developed alongside each other, and then tested on two very different imaging modalities.

Phase 1 established the dual-frequency feature fusion framework on top of a 9-channel spectral representation built from the log-magnitude and sine/cosine-encoded phase of each RGB channel. The centrepiece of this phase is the DFFF module, which fuses spatial and spectral activations via channel attention with learnable, per-domain weights; explainability is handled by a frequency-domain Score-CAM variant so that every prediction can be traced back to a region on the original scan.

Phase 2 lifted that single-scale design into a hierarchical, multi-scale architecture with three moving parts:

- Residual-style frequency gating at the intermediate backbone scales, specifically 28×28 (Layer 2) and 14×14 (Layer 3).
- A feature pyramid whose top-down fusion is controlled by a set of softmax-normalised logits learnt jointly with the rest of the network.
- A multi-scale Score-CAM that pulls frequency-aware attention from every level of the pyramid rather than from a single layer.

To show that MSPF is not tied to any one backbone, the architecture was evaluated across several families of CNNs. The best test accuracy on the Brain Tumor MRI dataset reached 85.63% with DenseNet-121, and the same model hit 84.45% on the Chest X-Ray Pneumonia dataset; ResNet-50, EfficientNet-B0 and MobileNetV2 all show consistent improvements over their own spatial-only baselines, which is what we would want from a backbone-agnostic extension. Alongside the accuracy gains, the multi-scale Score-CAM produces overlays that localise the relevant anatomy cleanly enough to support a clinical reading of each prediction.

7.2 Key Findings

1. **Spectral representations complement spatial features.** The 9-channel FFT encoding consistently outperforms the RGB baseline, and *encoding* the phase through sine and cosine rather than feeding the raw angle into the network makes training meaningfully more stable.
2. **Asymmetric multi-scale frequency processing is effective.** Pairing lightweight frequency-attention gates at the high-resolution intermediate layers with a full dual-domain fusion block at the coarsest layer gives the best trade-off between compute and expressive power; treating every scale identically would either be wasteful or unstable.
3. **Learnable pyramid fusion adapts to task complexity.** Because the softmax-normalised scale weights are learnt end-to-end, the network effectively decides for itself how much emphasis to put on each resolution, and the optimal balance is different on the brain-tumour and the pneumonia datasets.
4. **Robustness across CNN architectures.** MSPF carries over cleanly across ResNet, DenseNet, EfficientNet, MobileNet and VGG backbones; the specific numbers change with the backbone, but the direction of the improvement does not, which suggests the idea of layered spectral-plus-spatial fusion is the invariant here rather than any particular choice of architecture.
5. **Frequency-domain Score-CAM produces sharper attributions.** Running Score-CAM on the fused multi-scale features, and then mapping back to pixel space through the inverse FFT, yields saliency maps that follow anatomical boundaries more faithfully than the standard spatial-domain variant.

7.3 Future Works

1. **Multi-resolution frequency pyramid:** Run frequency analysis at three resolutions — 100×100 , 50×50 and 25×25 — and stitch the outputs together with lightweight CNN blocks and attention-based fusion.
2. **Spatial-frequency co-attention:** Let the spatial and frequency streams query each other bidirectionally rather than fuse one-shot; a cross-attention variant looks promising here, provided it can be done without inflating the parameter budget.
3. **Extended clinical validation:** Test the pipeline on larger, multi-institutional corpora — brain MRI, chest X-ray, retinal imaging and histopathology — with expert annotation, so that real-world utility and cross-domain robustness can be measured rather than assumed.
4. **Multi-scale CAM framework:** Build a dedicated multi-scale CAM pipeline in which every pyramid level carries its own frequency-aware interpretation so that a clinician can inspect the explanation at the spatial granularity that matters for a given case.
5. **Domain-adaptive frequency encoding:** Explore modality-specific spectral front-ends — wavelet-based decompositions for radiographs, for instance — that are better matched to the frequency characteristics of each imaging modality than a single, generic FFT lift.

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